

FROM CLASSROOM JOURNEY TO REAL WORLD INSIGHT



COFFEE SALES ANALYSIS USING GOOGLE SHEETS



DATA ANALYSIS



PIVOT TABLES



CHARTS



DASHBOARD



BUSINESS INSIGHTS

Prepared by:

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Order	Date	Coffee Name	Cash	Card	Money
001	01/03/24	Latte	70.00	150.00	220.00
002	02/03/24	Americano	50.00	130.00	130.00
003	03/03/24	Cappuccino	80.00	180.00	180.00
004	03/03/24	Mocha	60.00	140.00	200.00
005	03/03/24	Cappuccino	40.00	110.00	150.00
006	03/03/24	Americano	30.00	90.00	120.00
007	03/03/24	Cappuccino	20.00	80.00	120.00



A1 Date time

	A	B	C	D	E	F	G	H	I	J	K	L
1	Date time	Date	Payment Method	Card	Money	Coffee name	Day	Month	Weekday/ Week	Time Slot	Order Value	Category
2	2024-03-01 10:1	2024-03-01	card	ANON-0000-000	38.7	Latte	Friday	March	Weekday	Morning	Medium	
3	2024-03-01 12:1	2024-03-01	card	ANON-0000-000	38.7	Hot Chocolate	Friday	March	Weekday	Afternoon	Medium	
4	2024-03-01 12:2	2024-03-01	card	ANON-0000-000	38.7	Hot Chocolate	Friday	March	Weekday	Afternoon	Medium	
5	2024-03-01 13:4	2024-03-01	card	ANON-0000-000	28.9	Americano	Friday	March	Weekday	Afternoon	Low	
6	2024-03-01 13:4	2024-03-01	card	ANON-0000-000	38.7	Latte	Friday	March	Weekday	Afternoon	Medium	
7	2024-03-01 15:3	2024-03-01	card	ANON-0000-000	33.8	Americano with I	Friday	March	Weekday	Afternoon	Low	
8	2024-03-01 16:1	2024-03-01	card	ANON-0000-000	38.7	Hot Chocolate	Friday	March	Weekday	Afternoon	Medium	
9	2024-03-01 18:3	2024-03-01	card	ANON-0000-000	33.8	Americano with I	Friday	March	Weekday	Evening	Low	
10	2024-03-01 19:2	2024-03-01	card	ANON-0000-000	38.7	Cocoa	Friday	March	Weekday	Evening	Medium	
11	2024-03-01 19:2	2024-03-01	card	ANON-0000-000	33.8	Americano with I	Friday	March	Weekday	Evening	Low	
12	2024-03-01 19:2	2024-03-01	card	ANON-0000-000	33.8	Americano with I	Friday	March	Weekday	Evening	Low	
13	2024-03-02 10:2	2024-03-02	card	ANON-0000-000	28.9	Americano	Saturday	March	Weekend	Morning	Low	
14	2024-03-02 10:3	2024-03-02	cash		40	Latte	Saturday	March	Weekend	Morning	High	
15	2024-03-02 10:4	2024-03-02	card	ANON-0000-000	33.8	Americano with I	Saturday	March	Weekend	Morning	Low	
16	2024-03-02 11:5	2024-03-02	card	ANON-0000-000	33.8	Americano with I	Saturday	March	Weekend	Morning	Low	
17	2024-03-02 14:3	2024-03-02	card	ANON-0000-000	28.9	Americano	Saturday	March	Weekend	Afternoon	Low	
18	2024-03-02 16:3	2024-03-02	card	ANON-0000-000	33.8	Americano with I	Saturday	March	Weekend	Afternoon	Low	
19	2024-03-02 17:3	2024-03-02	card	ANON-0000-000	28.9	Americano	Saturday	March	Weekend	Evening	Low	
20	2024-03-03 10:1	2024-03-03	cash		40	Latte	Sunday	March	Weekend	Morning	High	
21	2024-03-03 10:2	2024-03-03	card	ANON-0000-000	38.7	Latte	Sunday	March	Weekend	Morning	Medium	
22	2024-03-03 11:3	2024-03-03	card	ANON-0000-000	28.9	Cortado	Sunday	March	Weekend	Morning	Low	
23	2024-03-03 12:2	2024-03-03	card	ANON-0000-000	28.9	Americano	Sunday	March	Weekend	Afternoon	Low	

Coffee Sales Data – From Raw to Analysis

RAW DATASET (SOURCE)

- Source real-world coffee shop sales data from the web — containing Date , Date & Time, Payment Method, Money and Coffee Name .
- Dataset had both cash & card transactions; card IDs were anonymized (ANON-XXXX) for data privacy compliance .
- Raw data was unstructured — no time segments, no behavioral tags, no value classification .

DATA ENHANCEMENT

- *Day & Month* — Extracted from datetime using TEXT() function .
- *Weekday / Weekend* — Classified using WEEKDAY() + IF() for behavioral segmentation .
- *Time Slot (Morning / Afternoon / Evening / Night)* — Derived via HOUR() + nested IF() .
- *Order Value Category (Low / Medium / High)* — Built using IFS() for revenue bucketing .

FORMATTING & PRESENTATION

- Applied **alternating row colors** for a clean, professional, easy-to-read layout .
- Applied visual hierarchy through color-coded headers — raw data vs. transformed columns .
- Dataset made fully structured and presentation-ready for pivot tables & charts .

KEY LEARNINGS

- Breaking down timestamps uncovered peak sales hours, weekday vs. weekend behavior & monthly trends .
- Google Sheets built-in functions (TEXT, IF, IFS, HOUR, WEEKDAY) are powerful enough for real-world EDA
- Clean, well-labelled data is the first step to any meaningful business insight or dashboard

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Business Questions & Key Insights :-

Questions	Solutions
What is the total revenue generated?	115432
What is the average order value?	32
What is the highest selling product?	Latte
Give peak sales hour.	22
Do customers spend more on weekends or weekdays?	Weekdays

Insights Interpretation :-

1. What is the total revenue generated?

Explanation:

Calculated the total revenue by summing all values in the Money column using the **SUM** function, resulting in a total revenue of **₹115,432**.

Formula: **=SUM('coffee sales'!E2:E)**

2. What is the average order value?

Explanation:

Calculated the **Average Order Value (AOV)** using the **AVERAGE** function, which returned **₹32**, indicating the average amount spent per customer transaction.

Formula: **=AVERAGE('coffee sales'!E:E)**

3. What is the highest selling product?

Explanation:

Used the **INDEX**, **MATCH**, and **MAX** functions to identify the product associated with the **highest sales value**. The analysis shows **Latte** as the highest-selling product.

Formula: **=INDEX('coffee sales'!F:F, MATCH(MAX('coffee sales'!E:E), 'coffee sales'!E:E, 0))**

4. Give peak sales hour.

Explanation:

Used the **MAX** function to identify the **peak sales hour** from the Hour column. The highest sales were recorded at **22:00 (10 PM)**.

Formula: **=MAX('coffee sales'!M:M)**

5. Do customers spend more on weekends or weekdays?

Explanation:

Compared the total revenue generated on **weekends** and **weekdays** using **SUMIF** and **IF** functions. The analysis shows that **weekdays generate higher sales** than weekends.

Formula: **=IF(SUMIF(I:I,"Weekend",E:E)>SUMIF(I:I,"Weekday",E:E),"Weekends","Weekdays")**

K11  

	A	B	C	D	E	F	G	
1								
2	Which coffee products generate the highest revenue for the business? Latte by ₹27866.3	Coffee name	SUM of Money		Which time slot generates maximum revenue? During Afternoon by ₹39018	Time Slot	SUM of Money	
3		Americano	15062			Afternoon	39018	
4		Americano with Milk	25269			Evening	28820	
5		Cappuccino	18034			Morning	37380	
6		Cocoa	8678			Night	10213	
7		Cortado	7535			Grand Total	115432	
8		Espresso	2814					
9		Hot Chocolate	10172			Find all the coffee products that are greater than 35 sold on weekends .	COUNTA of Money	Weekday/ Weekend
10		Latte	27866				Coffee name	Weekend
11	Grand Total	115432		Cappuccino	110			
12				Cocoa	63			
13	Which payment method is most preferred by customers?	Payment Method	SUM of Money	Hot Chocolate	79			
14	It's card	card	112246		Latte	188		
15		cash	3186		Grand Total	440		
16		Grand Total	115432					
17								
18	Do customers spend more on weekends or weekdays?	Weekday/ Weekend	SUM of Money					
19	During weekdays	Weekday	82114.56					
20		Weekend	33317					

Pivot Table Findings

Q1 · Which coffee product generates the highest revenue?

Ans – I've inserted a pivot table with Coffee Name in rows and SUM of Money in values. Looking at all products, Latte gave the highest revenue of **₹27,866** so, it is highlighted.

Q2 · Which time slot generates maximum revenue?

Ans – I've inserted a pivot table with Time Slot in rows and SUM of Money in values. Comparing all four slots — Afternoon, Morning, Evening and Night — Afternoon generated the maximum revenue of **₹39,018** so, it is highlighted.

Q3 · Which payment method is most preferred by customers?

Ans – I've inserted a pivot table with Payment Method in rows and SUM of Money in values. Card payments totalled **₹1,12,246** versus cash at only **₹3,186** making Card the most preferred method — highlighted accordingly.

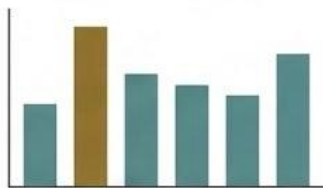
Q4 · Do customers spend more on weekends or weekdays?

Ans – I've inserted a pivot table with Weekday/Weekend in rows and SUM of Money in values. Weekdays showed ₹82,114 against weekends at ₹33,317, so customers clearly spend more on weekdays — highlighted in the table.

Q5 · Find all coffee products sold more than 35 units on weekends

Ans – I've inserted a pivot table with Coffee Name in rows, Weekday/Weekend as a filter (set to Weekend), and COUNTA of Money in values. After applying a value filter for greater than 35 four products remained — Latte (188), Cappuccino (110), Hot Chocolate (79), and Cocoa (63) — all highlighted as they cross the 35-unit threshold.

Pivot Table Findings



Highest Revenue Product?

Latte: ₹27,866



Maximum Revenue Time Slot?

Afternoon: ₹39,018



Most Preferred Payment Method?

Card: ₹1,12,246



More Spent on Weekends/Weekdays?

Weekdays: ₹82,114



Products > 35 Units Sold on Weekends

Latte (188), Cappuccino (110), Hot Chocolate (79), Cocoa (63)

Chart Analysis & Insights

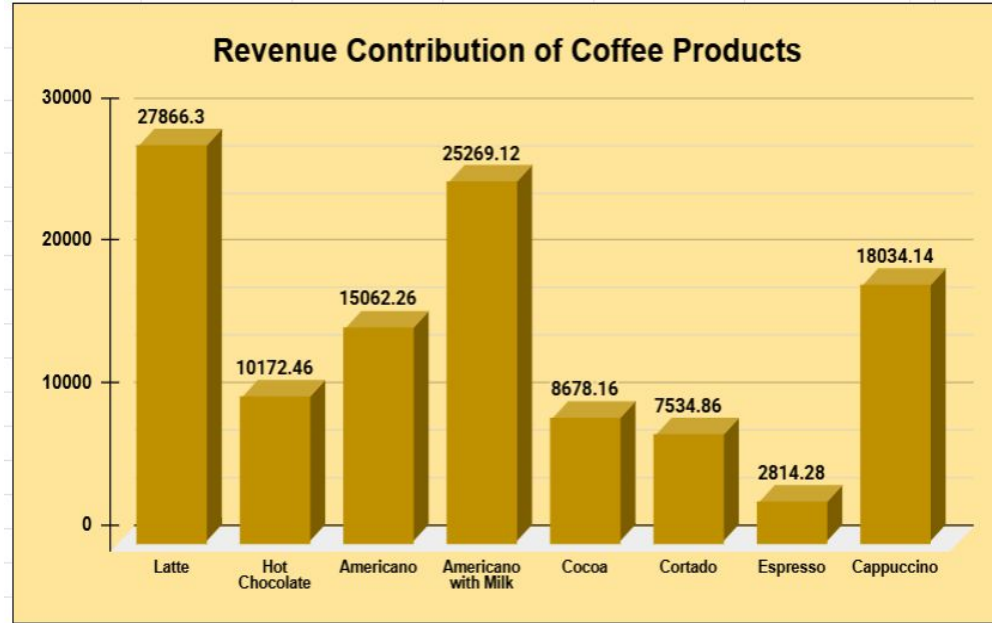
Q. 1 Which coffee product generates the highest revenue?

Explanation:

- i) This column chart compares total revenue across 8 coffee products.
- ii) The X-axis shows product names .
- iii) The Y-axis represents revenue ₹0–₹30,000, **Latte** leads at **₹27,866.3**, followed by **Americano with Milk** at **₹25,269.12** while **Espresso** records the lowest at **₹2,814.28** .
- iv) Highlighting a significant performance gap across the product portfolio .

Procedure:

Insert → Chart → Column Chart → Select product & revenue data → Enable data labels on bars → Add title & format axes .



Q. How does revenue change month by month?

Explanation:

i) This line chart tracks revenue movement across 12 months (Mar–Feb).

ii) The X-axis represents months and the Y-axis shows revenue (₹0–₹20,000).

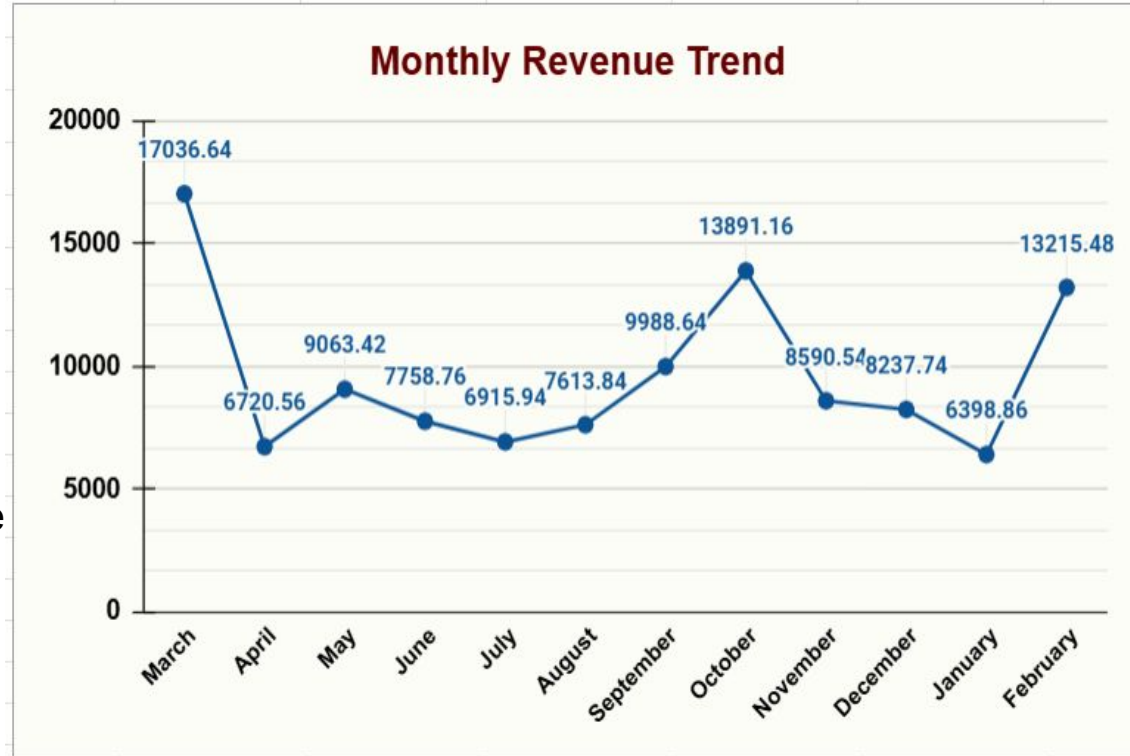
iii) **March** peaked at **₹17,036.64** followed by a sharp drop to **₹6,720.56** in **April**.

iv) Revenue recovered in **October** in **₹13,891.16** and closed strong in

v) **February** in **₹13,215.48** indicating a cyclical revenue pattern with two notable peaks.

Procedure:

Insert → Chart → Line Chart → Select month & revenue data → Enable data labels at each point → Add title & format axes.



Q. Which payment method contributes the most to revenue?

Explanation:

i) This pie chart illustrates the proportional split of revenue between payment methods used by customers.

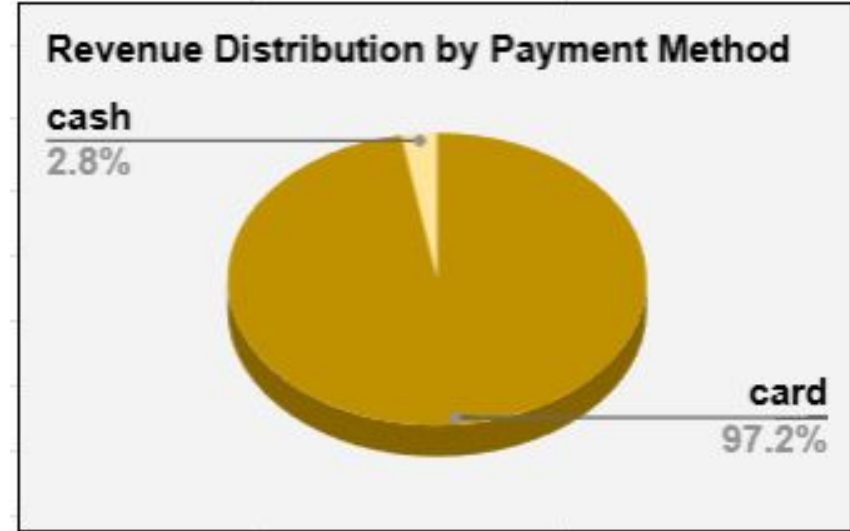
Card payments dominate accounting for **97.2%** of **total revenue**.

ii) While **cash transactions** contribute a marginal **2.8%**.

iii) This indicates a strong customer preference for digital/card-based payments, reflecting broader trends toward cashless transactions.

Procedure:

Insert → Chart → Pie Chart → Select payment method & revenue data → Enable data labels (percentage) → Add title & legend.



Q. How does customer spending vary across different time slots?

Explanation:

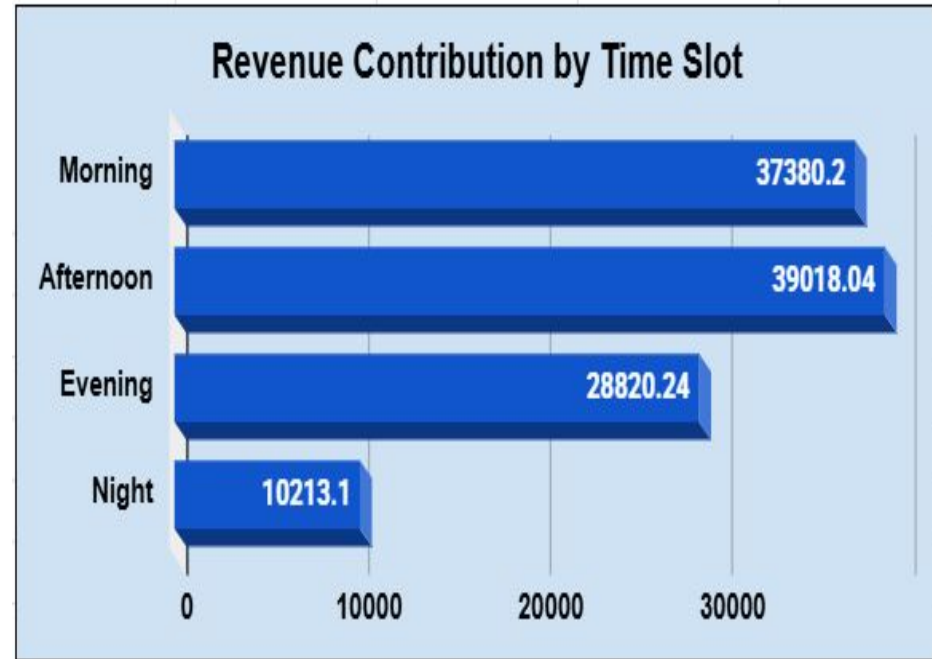
i) This bar chart shows the revenue generated during different time slots of the day.

ii) **Afternoon** has the highest revenue (**₹39,018.04**), followed by **Morning** (**₹37,380.20**) and **Evening** (**₹28,820.24**).

iii) **Night** contributes the lowest revenue (**₹10,213.10**), indicating lower customer activity.

Procedure:

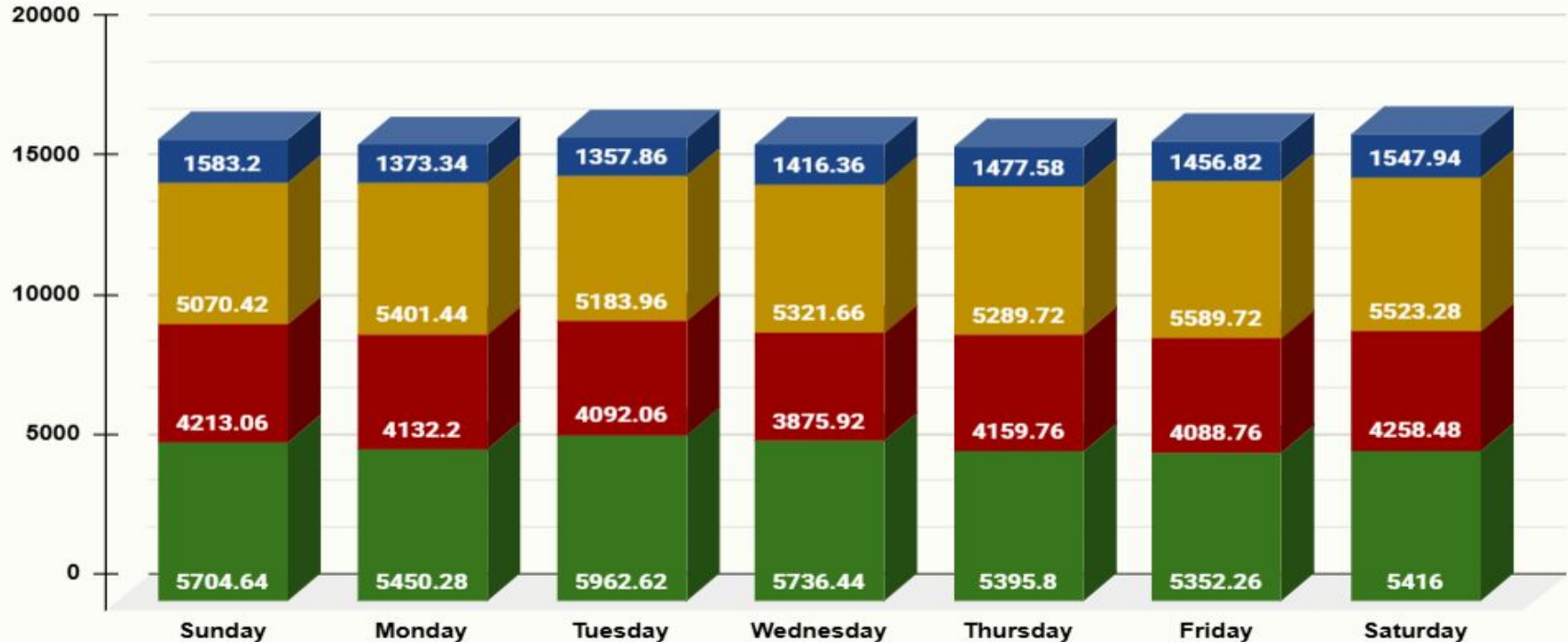
Insert → Chart → Bar Chart → Select Time Slot & Revenue data → Enable Data Labels → Add Chart Title.



Q. How much revenue is generated during each time slot on different days?

Daily Revenue Analysis by Time Slot

■ Night ■ Morning ■ Evening ■ Afternoon



Q. How much revenue is generated during each time slot on different days?

Explanation:

i) This stacked column chart shows the daily revenue distribution across four time slots: **Morning, Afternoon, Evening, and Night**.

ii) **Morning** and **Afternoon** contribute the highest share of revenue on most days, while **Night** records the lowest revenue.

iii) The chart helps compare customer purchasing patterns throughout the week and identifies the busiest time slots for each day.

Procedure:

Insert → Chart → Stacked Column Chart → Select Day and Time Slot Revenue data → Enable Data Labels → Add Chart Title & Legend.

Conclusion :-

- Analyzed the coffee sales dataset using **Google Sheets** to derive actionable business insights.
- Applied functions such as **SUM, AVERAGE, INDEX, MATCH, MAX, IF, and SUMIF** to answer business questions.
- Created **Pivot Tables** and **interactive charts** to visualize sales trends, customer preferences, and product performance.
- Identified key findings such as **Latte being the top revenue-generating product, Afternoon as the highest revenue time slot, Card as the preferred payment method, and Weekdays generating higher sales than weekends.**
- This project strengthened my skills in **data analysis, reporting, visualization, and business decision-making using Google Sheets.**

Thank You..!!